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Topic:

Monitoring Concentration in ADHD Youth: An EEG and Data Science Approach

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General Introduction

In Tunisia, and many societies, neurophysiological issues in children are often underdiagnosed and inadequately monitored over the long term, leading to gaps in effective management and support.
Among these, Attention Deficit Hyperactivity Disorder (ADHD) is a prevalent neurodevelopmental disorder characterized by symptoms of inattention, hyperactivity, and impulsivity.

Globally, it is estimated that ADHD affects approximately 5-7% [1] of children and adolescents. In Tunisia, ADHD prevalence rates among school-aged children are reported to be in line with global statistics, affecting around 6-8% [2] of the youth population.

Children with ADHD often face challenges in their educational and social interactions, which can impact their future confidence and success. Without proper diagnosis and intervention, these challenges can hinder their performance in school, affect student-teacher relationships, and strain parent-child interactions at home. ADHD is a neurobehavioral disorder that can be managed and, in some cases, significantly improved with the right strategies and interventions. [3]

One effective approach to managing ADHD involves tasks that require full concentration for short periods, such as a 15-minute focused activity, which can help stimulate neural connections. To support this, we have developed a robotic system designed to follow the child's routine, particularly during study periods. Children with ADHD often struggle to stay still and focus, especially during homework time. Our goal is to implement a portable device that senses and monitors the child's concentration levels during study sessions at home.

The system works by rewarding the child with positive feedback from the robot for every 15 minutes of sustained concentration. This approach aims to encourage the child psychologically to improve focus and productivity on their homework. Additionally, parents can monitor their child's concentration levels through a mobile application, receiving alerts if their child is not concentrating on the task. This real-time feedback allows parents to intervene and implement different strategies to enhance their child's focus and engagement in future tasks.

1 Project Context:

1.1 Comparative Study:

Multiple biofeedback methods are used to assess an individual's mental status based on their physiological responses [4]. These methods include:

- **Galvanic Skin Response (GSR):** Also known as skin conductance, this method measures the amount of sweat on the skin's surface, serving as a useful marker for detecting levels of emotional arousal.
- **Electromyography (EMG):** Sensors placed at various points on the body detect changes in muscle tension over time by monitoring electrical activity resulting from muscle contractions.

While these biofeedback methods can provide insights into a person's concentration, they are inaccurate. Various factors, such as an individual's health status, seasonal changes, and differences in physical activity levels, can influence the readings. Additionally, wearing multiple sensors may be uncomfortable, especially for children.

Apart from biofeedback, another research approach involves using multiple cameras to measure concentration [5]. However, privacy concerns have become a significant issue, with widespread debates and legal implications. As a result, having cameras installed throughout a home can be intimidating for parents and children alike.

1.2 Adopted Solution:

To address these challenges, connecting directly with the brain and monitoring the signals that indicate different mental states is essential. Recent neurotechnology and data science advancements offer promising new avenues for objective assessment. Electroencephalography (EEG), a non-invasive method that records brain activity, has shown potential in detecting concentration levels through brain wave analysis.

By applying data science techniques to EEG signals, using Python, robust models can be developed to objectively monitor and assess the attention states of individuals with ADHD.

An example of such a portable EEG solution is the Emotiv product Cutting-Edge EEG Solutions, which represent different EEG Headsets with various channel numbers.

1.3 Limitations:

In this project, we faced several limitations. One significant challenge was the high cost of portable EEG devices. Additionally, there were time constraints related to building and integrating the robot that interacts with the EEG device. These limitations highlight areas for future improvement and refinement in making the technology more accessible and fully realized.

2 Modelisation & Analysis

2.1 EEG Mechanism:

EEG is a reliable method that provides information about the brain's background activity and reflects the underlying processes of cognition and behavior, as shown in Table 1. [6]

Table 1. EEG sub-bands are associated with different brain functions

Sub-Band	Frequency Range	Associated Brain Function
Delta	0.5–4 Hz	Deep sleep or unconsciousness
Theta	4–8 Hz	Sleep or drowsiness and recall
Alpha	8–13 Hz	Eye closed and visual stimuli are limited
Beta	13–30 Hz	Attentive to stimuli or problem-solving
Gamma	30 Hz and above	Movement, emotional processing, and high-level mental activity

What is special about children with ADHD is that they often exhibit a common brain-wave pattern characterized by an abundance of slow (delta or theta) brain waves and a shortage of fast (beta) brain waves. These brainwaves are read through electrodes fixed at specific points on the scalp, each corresponding to a particular brain region. To target areas responsible for concentration and attention, our configuration includes electrodes covering the frontal, central, and parietal regions, which are commonly implicated in ADHD and attentional processes.

This is a 10–20 system for electrode position with A1 and A2 reference electrodes [7]

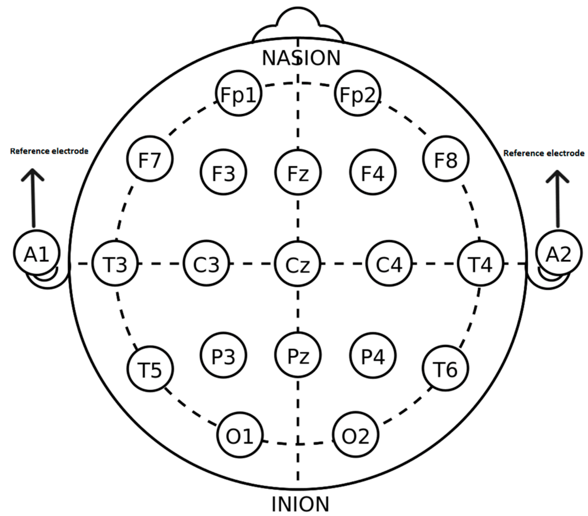


Figure 1

from which we will choose specific electrodes to monitor theta and beta waves:

- **Fz:** Midline frontal electrode (for monitoring both theta and beta activity in the frontal region, important for attention and executive function)
- **Cz**: Midline central electrode (for monitoring central beta activity, which is associated with alertness and concentration)
- **F3, F4:** Frontal electrodes on the left and right sides (for monitoring frontal beta activity, important for cognitive control and attention)
- **C3, C4:** Central electrodes on the left and right sides (for monitoring central beta activity)
- **Pz:** Midline parietal electrode (for capturing posterior theta activity, which may be relevant for attention and cognitive processing)

If the beta waves, which range from 13 to 21 cycles per second, appear continuously for 15 minutes, it indicates that the child is concentrating. This triggers a rewarding reaction from a linked robot. Conversely, if the readings drop to 4 to 8 cycles per second, indicating the appearance of theta waves, it means the child is no longer concentrating. In such cases, a notification is sent so that parents can monitor their child's focus levels.

2.2 Data Acquisation:

2.2.1 Data Description:

EEG brain recordings are freely available at the IEEE DataPort. The EEG data clips of ADHD recordings were performed based on an experiment where a set of pictures of cartoon characters was shown to the children, who were asked to count the characters [8]. Each data clip contains a matrix of EEG sample values arranged in a row × column format as electrode channel × time.

In our subject's data set, 19 electrode channels were sampled at 128 Hz. For our analysis, we will focus on the Fz, Cz, Pz, C3, F3, C4, and F4 labels, which correspond to channels 17, 18, 19, 5, 3, 6, and 4 in the data, respectively. We will extract a 30-minute segment from the available 3 hours of data clips for our analysis.

Our dataset includes recordings from 61 children. We will select a random child's recording and process it through our Python program to analyze the EEG signals and assess concentration levels.

2.2.2 Data Preprocessing:

To apply the most accurate read of our data, we are following the key steps of data preprocessing as highlighted in a project paper by our colleagues at Stanford University. This involves ensuring the quality and consistency of the data, which is crucial for reliable analysis and results [9].

2.2.2.1 Normalization:

Since the range of voltage readings in the EEG dataset varies widely, we normalized the data to avoid heavily weighting readings that are very positive or very negative. For each segment, we calculated the minimum voltage reading over all 15 channels, as well as the range of the data for that segment. We normalized each EEG reading x, by computing:

$$x_{normalized} = \frac{x - min(x)}{|max(x) - min(x)|}$$

2.2.2.2 Filtering Data:

To ensure the accuracy of our EEG signal analysis, we apply **a bandpass filter to the data**. This process involves filtering the data to only allow frequencies within a specified range (the frequency band) to pass through, effectively removing noise and irrelevant frequencies. The bandpass filter is implemented **using the Nyquist filter**, which is appropriate for digital signal processing.

The process begins by defining the Nyquist frequency, which is half of the sampling frequency (*f*s). The desired low and high cutoff frequencies are then divided by the Nyquist frequency to normalize them. **A Butterworth filter**, known for its smooth frequency response, is designed with the specified order and cutoff frequencies. The filter coefficients are computed, and the filter is applied to the data, resulting in filtered data that retains only the frequencies within the specified band.

In addition to the bandpass filter, we will implement another filtering technique to **address the eye blink effect** (it appears in position variation superior to min 3*variation average in the signal), which can significantly distort our EEG signal.

This effect often manifests as a substantial amplification in the signal, making it necessary to mitigate its impact for accurate analysis. To achieve this, we will integrate specific filter-targeting positions where variations exceed three times the average variation in the signal. This filter aims to identify and reconstruct artifacts caused by eye blinks, ensuring that the resulting EEG data is free from such distortions. By combining these filtering methods, we aim to enhance the quality and reliability of our EEG data, enabling more precise analysis of concentration levels and attention states.

2.2.3 Feature Extraction:

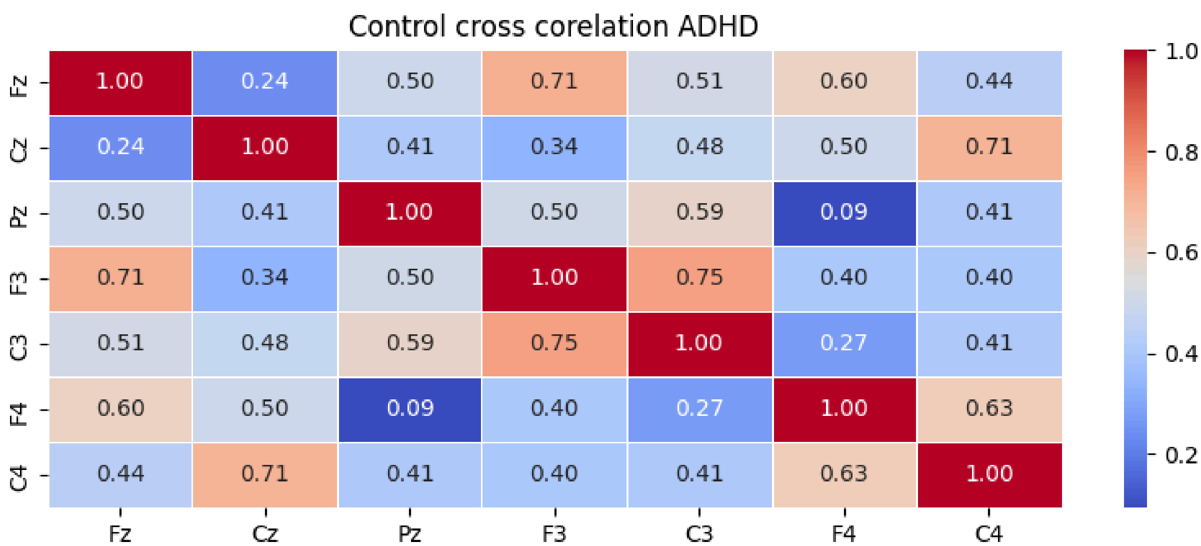
We move forward with the feature extractor to choose the necessary variables for our classification, by following the steps:

Mean and Standard Deviation: These fundamental statistical measures provide us with insights into the overall signal amplitude and variability. By examining the mean and standard deviation, we understand the general characteristics of the EEG signals.

Power Spectral Density (PSD): This measure evaluates the signal's power at various frequencies, in our case we utilized the Fourier Transform. This technique is essential for understanding how power is distributed across different frequency components of the EEG signals.

Band Power: Now, we focus on the power within specific frequency bands, such as Delta (0.5-4 Hz), Theta (4-8 Hz), Alpha (8-13 Hz), Beta (13-30 Hz), and Gamma (30-100 Hz). To identify and isolate the contributions of different frequency ranges to the overall EEG signal.

Cross-correlation: This measure assesses the similarity between EEG signals from different electrodes in the frequency domain. Cross-correlation is useful for understanding the relationships and interactions between various brain regions.



By applying these feature extraction techniques, we built a more comprehensive feature space, to a better understanding of our classifier..

2.2.3 Data Classification:

Having a clear understanding of our data features, we move forward with classifying our EEG data based on power analysis. Using the *'welch'* method from the *'scipy.signal' library*, we analyze the power spectral density (PSD) of the EEG signals. The process involves segmenting the data into one-second intervals and calculating the power within specific frequency bands for each electrode.

The electrodes we focus on include Fz, Cz, Pz, F3, C3, F4, and C4. For each one-second segment, we compute the power of the beta (12-30 Hz) and theta (4-8 Hz) waves. The *'welch'* method provides the frequency and PSD values, from which we derive the power within these frequency ranges.

The results for each segment and electrode are then organized into a structured format, creating a detailed data frame that captures each second and electrode's beta and theta power. This data frame serves as the foundation for further analysis and classification, allowing us to identify patterns that indicate concentration levels.

By focusing on the power within these specific frequency bands, we gain valuable insights into the cognitive states of individuals. This facilitates more accurate classification of their concentration levels, enabling us to build a reliable model for monitoring and assessing the attention states of individuals with ADHD.

3 Conclusion:

In this project, we implemented a supervised binary classifier for EEG recordings. Throughout our work, we have been able to explore our data processing method and extract a feature that we used to extract the beta-to-theta ratio on our data and classify the patient's state for each second.

This classifier will return to us if the individual is concentrated or not, the continuousness of a specific return for 15 minutes will allow us to send that message to our Robot in the future and display the according message.

To further explore we still need to verify the accuracy of our model using a non-linear classification method.

4 Future Work:

4.1 Hardware:

- **Robot Conception:** Initiate the full conceptualization and design process for our robot. This involves defining the physical appearance, functionality, and features of the robot, ensuring it aligns with the objectives of our project and meets the needs of our target users.
- **IoT Integration with ESP**: Integrate IoT technology, specifically using ESP (e.g., ESP32 or ESP8266), to enhance the responsiveness of our robot. This integration will enable the robot to receive and respond to notifications in real time.

4.2 Software:

- **Application Interface Development**: Develop the user interface for our application to seamlessly integrate with our backend code. The interface will provide parents with access to various statistics and insights derived from our product. This includes real-time performance metrics of their child during study periods, which can be viewed on a daily, weekly, and monthly basis. Additionally, the interface will display progress made within specific time frames, allowing parents to track their child's development over time.

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